

# Multimodal Deepfake Speech Detection

*Audio (wav2vec2) and audio-visual lip-sync detection on FakeAVCeleb v1.2, with leakage analysis, cross-generator generalization, and robustness*

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**Abstract:** We study the detection of AI-generated (synthetic) speech using a fine-tuned wav2vec2 audio classifier and a complementary audio-visual lip-sync stream, on the FakeAVCeleb v1.2 dataset. A baseline audio model reaches a near-perfect in-distribution Equal Error Rate (EER) of 0.000, which we show to be misleading. First, we uncover a hidden data leak (64% of the audio is duplicated across clips), yet removing it barely changes the score, proving the in-distribution task is trivially separable. Second, a cross-generator generalization test reveals the model collapses on speech from an unseen generator (EER 0.37), indicating it learned a generator-specific fingerprint rather than fakeness. Third, an audio-visual sync feature is generator-agnostic and isolates the dubbing case the audio model is blind to, demonstrating complementary failure modes. Finally, augmentation and multi-generator training partially close the generalization gap. Our contribution is a rigorous, reproducible evaluation methodology rather than a headline accuracy.

## 1. Introduction

Synthetic-speech generators have reached a quality at which reliable detection is a genuine security requirement: cloned voices can drive fraud, disinformation, and impersonation. The dominant detection recipe fine-tunes a pretrained self-supervised speech model on a spoof-detection corpus. We follow this recipe on FakeAVCeleb, but we treat the evaluation itself and not the headline accuracy as the main contributor.

Our central message is cautionary. A wav2vec2 detector trained on FakeAVCeleb attains an in-distribution EER of essentially zero. We demonstrate through three independent analyses that this figure does not describe a usable detector. The contributions are:

- A diagnosis of a subtle audio-reuse data leak in FakeAVCeleb, and a hash-based correction.
- Evidence that the leak is not the cause of the perfect score and the in-distribution task is trivially separable irrespective of the split.
- A cross-generator generalization study showing the model fails on an unseen generator.

- A generator-agnostic audio-visual sync stream with complementary failure modes, and a late-fusion ablation.
- A robustness extension (augmentation, multi-generator training) that partially recovers cross-generator performance.

## 2. Related Work

Anti-spoofing for automatic speaker verification is benchmarked by the ASVspoof challenges, which popularized the EER metric and exposed the difficulty of generalizing across unseen attacks. Self-supervised speech models such as wav2vec2 provide strong front-ends that, when fine-tuned, achieve low in-distribution EER on these benchmarks. FakeAVCeleb extends the setting to audio-visual deepfakes, combining face manipulation (Faceswap, FSGAN, Wav2Lip) with voice cloning (SV2TTS). Prior audio-visual deepfake work often targets the video. We instead focus on the speech and use the visual channel only through lip-sync, which is independent of the audio generator. The well-documented fragility of spoof detectors to out-of-distribution generators motivates our generalization study and robustness extension.

### 3. Background

#### 3.1 Synthetic speech and artifacts

Synthetic speech is produced by text-to-speech or voice-cloning systems. Detectors exploit artifacts, spectral and phase fingerprints introduced by the synthesis pipeline, especially the vocoder, that are absent in genuine recordings. FakeAVCeleb fake audio comes from SV2TTS, a three-stage cloning system: a speaker encoder embeds the target voice, a Tacotron-style synthesizer produces a mel-spectrogram from text and embedding, and a WaveRNN-style vocoder reconstructs the waveform. Coqui XTTS v2, used here as an out-of-distribution generator, has a different architecture and therefore a different artifact signature.

#### 3.2 wav2vec2 and self-supervised learning

wav2vec2 learns speech representations without labels: a convolutional feature encoder maps the raw waveform to local features (~one vector per 20 ms), a Transformer adds context, and a contrastive objective over masked, quantized latents drives pretraining. The result transfers well to downstream speech tasks with limited labelled data.

#### 3.3 Transfer learning and fine-tuning

We take the pretrained model and attach a two-class head. We freeze the convolutional encoder (its low-level filters are already general, which curbs overfitting and saves compute) and fine-tune the Transformer and head. Mixed-precision training and a class-weighted loss handle the 8 GB GPU budget and the mild class imbalance respectively.

#### 3.4 Evaluation metrics

The Equal Error Rate (EER) is the threshold-independent operating point at which the false-accept rate equals the false-reject rate. It is the ASVspoof standard, with 0 perfect and 0.5 chance. We complement it with AUC (ranking quality), accuracy, a confusion matrix, and a per-category

(A/B/C/D) breakdown where the last being our key diagnostic for shortcut learning.

#### 3.5 Leakage, domain shift, and shortcuts

Data leakage occurs when test information reaches training, inflating apparent performance. The canonical case is the same (or near-identical) sample appearing in both splits, enabling memorization. Distribution (domain) shift is a systematic difference between training and deployment data which here is, a different generator or recording channel. Shortcut learning is the tendency to exploit an easy but non-causal correlation: if real clips are compressed YouTube audio and fakes are clean synthesis, a model may separate by recording channel rather than fakeness. Crucially, an in-distribution test cannot detect such a shortcut but only out-of-distribution data can.

### 4. Dataset

FakeAVCeleb v1.2 contains 21,544 clips across four categories defined by whether the video and the audio are real or fake. Because this is a speech task, the label is the audio truth, not the video.

Table 1. FakeAVCeleb categories; the label is the audio truth.

| Category            | Video | Audio | Label | Clips  |
|---------------------|-------|-------|-------|--------|
| RealVideo-RealAudio | real  | real  | REAL  | 500    |
| FakeVideo-RealAudio | fake  | real  | REAL  | 9,709  |
| RealVideo-FakeAudio | real  | fake  | FAKE  | 500    |
| FakeVideo-FakeAudio | fake  | fake  | FAKE  | 10,835 |

All 500 speakers appear in both classes, so a naive random split would leak voices. The classes are mildly imbalanced (~53% fake), handled by a class-weighted loss. Fake audio is produced by a single generator (SV2TTS), which becomes important for generalization.

## 5. Experimental Setup

All experiments run on a single NVIDIA RTX 3070 (8 GB), with PyTorch (CUDA 12.1) and Hugging Face Transformers. Audio is extracted with ffmpeg, the visual stream uses MediaPipe and the out-of-distribution generator (XTTS) runs in an isolated environment to avoid dependency conflicts.

Table 2. Training configuration.

| Hyperparameter | Value                        |
|----------------|------------------------------|
| Backbone       | facebook/wav2vec2-base       |
| Frozen         | CNN feature encoder          |
| Optimizer      | AdamW, lr 2e-5               |
| Precision      | fp16 (mixed)                 |
| Batch size     | 8                            |
| Epochs         | 3                            |
| Loss           | class-weighted cross-entropy |
| Clip length    | 4 s @ 16 kHz mono            |
| Selection      | best validation EER          |

## 6. Methodology

### 6.1 Indexing and extraction

We index all clips and labels by globbing the file tree (robust to inconsistent metadata), then extract mono 16 kHz audio with ffmpeg into a cache. Clips are cropped to four seconds (random crop in training, center crop in evaluation).

### 6.2 Splitting

Because every speaker spans both classes, we first split by speaker (70/15/15) so all clips of a voice stay together, asserting zero speaker overlap. Section 8 shows why this is necessary but not sufficient for audio.

### 6.3 Model and training

We fine-tune wav2vec2-base with the configuration in Table 2, selecting the checkpoint with the lowest validation EER.

### 6.4 Evaluation protocol

We report EER, AUC, accuracy, a confusion matrix, and a per-category breakdown on the held-out test speakers, and separately on out-of-distribution data (Section 9).

## 7. Baseline Results

On the speaker-disjoint split the model reaches test EER 0.0009, AUC 1.000, with all four categories near 100% accuracy and a confusion matrix of essentially zero errors. Near-perfect numbers are reported in the literature for such data, so rather than accept the result we tried to dig further and investigate.

## 8. A Hidden Data Leak

Diagnostics showed that, although train/test speaker overlap was zero, the same audio waveform appeared on both sides: of 18,388 train+test clips, only 6,807 audio hashes were distinct and 892 hashes occurred in both splits.

The cause is structural: FakeAVCeleb reuses one cloned voice across many target faces. The folder identity is the video/target identity, but the voice identity is encoded in the filename. Splitting by face keeps faces disjoint while letting voices and exact waveforms cross the split.

A filename-parsing fix failed because the voice identity is encoded differently per generation method. The robust fix groups by audio content hash and deduplicates: 21,544 clips reduce to 7,752 unique audios (64% were duplicates), after which cross-split hashes are zero.

Table 3. Leaky vs clean split; the fix barely changes the score.

| Split flow            | Train/Test     | Test EER | AUC   |
|-----------------------|----------------|----------|-------|
| Leaky (by face id)    | 15,088 / 3,300 | 0.0009   | 1.000 |
| Clean (by audio hash) | 5,426 / 1,165  | 0.000    | 1.000 |

**Key insight:** removing the leak and 64% of duplicate audio did not reduce the in-distribution score (0.0009 to 0.000). The leak was real but not

the cause. The task is trivially separable in-distribution, pointing to a channel/generator shortcut that no split can fix.

## 9. Cross-Generator Generalization

Since the in-distribution EER is  $\sim 0$  regardless of split, only out-of-distribution data can distinguish a real detector from a shortcut. We evaluate the clean model on 200 LibriSpeech clips (genuine, clean studio audio) and 100 Coqui XTTS v2 clips (a different generator), cloning the LibriSpeech voices.

Table 4. Out-of-distribution performance of the clean model.

| Set                | n   | Mean P(fake) | Accuracy               |
|--------------------|-----|--------------|------------------------|
| REAL (LibriSpeech) | 200 | 0.187        | 0.815                  |
| FAKE (XTTS v2)     | 100 | 0.455        | 0.440                  |
| Combined           | 300 | —            | AUC 0.710 / EER 0.3725 |

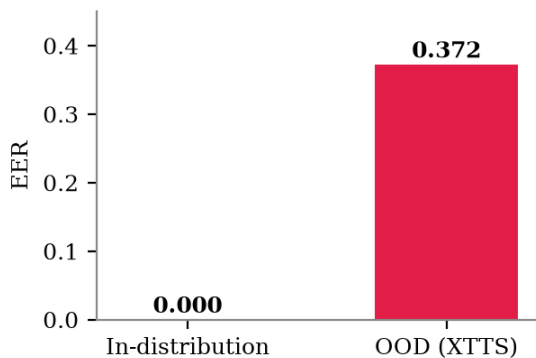


Figure 1. In-distribution vs out-of-distribution EER. A perfect 0.000 collapses to 0.37 on an unseen generator.

The model is not merely a channel detector where 81.5% of clean LibriSpeech reals survive but it fails on novel fakes, catching only 44% of XTTS. It learned the SV2TTS fingerprint, not fakeness. This is the canonical generalization failure of audio-deepfake detectors, and it is invisible to any in-distribution evaluation.

## 10. Multimodal: Audio-Visual Sync

The visual stream measures lip-sync: MediaPipe mouth-openness per frame is cross-correlated with the audio RMS envelope over small lags. Genuine speech and Wav2Lip-resynced fakes are well synced, audio dubbed onto an unmodified face is not. The feature is generator-agnostic, so it can catch fakes the audio model misses.

Table 5. Sync correlation by category, sync isolates dubbing.

| Category                       | Sync corr. | Note                         |
|--------------------------------|------------|------------------------------|
| FakeVideo-FakeAudio (Wav2Lip)  | 0.249      | lips re-synced to fake audio |
| RealVideo-RealAudio (genuine)  | 0.229      | naturally synced             |
| FakeVideo-RealAudio (faceswap) | 0.172      | medium                       |
| RealVideo-FakeAudio (dubbed)   | 0.110      | lowest: real lips mismatch   |

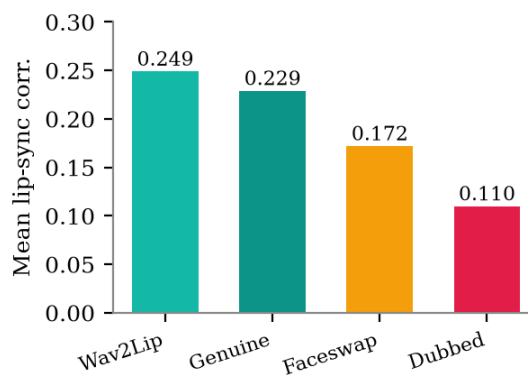


Figure 2. Lip-sync correlation by category. The dubbed case (real face, fake audio) has the lowest synchrony.

Table 6. Fusion ablation (160 clips).

| Model              | Acc   | AUC   | EER   |
|--------------------|-------|-------|-------|
| Audio-only         | 1.000 | 1.000 | 0.000 |
| Visual-only (sync) | 0.487 | 0.541 | 0.419 |
| Fused              | 1.000 | 1.000 | 0.000 |

Audio saturates in-distribution, so fusion adds no in-distribution gain which is an honest result. Visual sync is a weak global classifier but a meaningful dubbing detector (AUC 0.70 on real-face clips). The modalities fail on different cases: audio catches synthesis but not new generators; sync catches dubbing but not Wav2Lip resync. The multimodal value is robustness through complementary failure modes, not additive accuracy.

## 11. Robustness Extension

To mitigate the generalization failure we train with on-the-fly augmentation (noise, gain, codec simulation, band-limiting, reverb) and, optionally, multi-generator data (held-in XTTS and LibriSpeech), evaluating on a disjoint held-out OOD set.

Table 7. Robustness results on held-out OOD data.

| Model            | Aug | acc(real) | acc(fake) | AUC   | EER    |
|------------------|-----|-----------|-----------|-------|--------|
| clean (baseline) | no  | 0.815     | 0.440     | 0.710 | 0.3725 |
| robust_aug       | yes | 0.830     | 0.560     | 0.782 | 0.300  |
| robust_multigen  | yes | 1.000     | 0.960     | 1.000 | 0.000  |

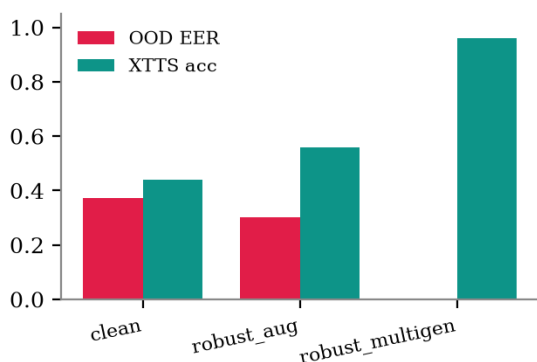


Figure 3. OOD EER (lower better) and XTTS detection accuracy (higher better) across the three models.

Augmentation alone with no XTTS in training lifts XTTS detection from below chance (0.44) to 0.56 and OOD EER from 0.3725 to 0.300,

confirming a real, breakable channel shortcut. Adding XTTS to training drives OOD EER to  $\sim 0$ , with the caveat that the held-out XTTS shares its generator with training (within-generator generalization, not true leave-one-generator-out).

## 12. Discussion

A perfect in-distribution metric can be nearly meaningless: EER 0.000 coexisted with 0.37 on an unseen generator. Rigorous evaluation which contains leakage analysis, per-category breakdowns, and out-of-distribution tests changed our understanding where accuracy alone did not. The leak was real but not the cause of the perfect score. Multimodality is best framed as complementary failure modes rather than additive accuracy. Generalization is partly fixable: augmentation helps across generators, and diverse training generators help more.

## 13. Ethical Considerations

Deepfake-speech detection is a defensive technology, but reporting inflated in-distribution numbers can create false confidence and real-world harm when a deployed detector silently fails on a new generator. Our results argue for routine out-of-distribution and cross-generator reporting before any deployment claim. The cloning tools used here are employed solely to evaluate detection, on public research data.

## 14. Limitations

FakeAVCeleb uses a single audio generator and our OOD adds only one more, so true leave-one-generator-out is out of scope. The visual sync is a hand-built feature rather than a trained network, and its absolute correlations are low (though the relative ordering is correct). The visual evaluation used 160 clips. The multi-generator test shares its generator with training.

## 15. Future Work

Three or more generators would enable genuine leave-one-generator-out evaluation. A trained audio-visual synchrony network with feature-level

fusion would strengthen the visual stream. A larger out-of-distribution visual evaluation requires OOD video, not only audio and stronger augmentation (real codecs and measured room impulse responses) are natural next steps.

## 16. Conclusion

We trained and tested a wav2vec2 fake-speech detector that is near-perfect in distribution, then showed why that number is a mirage: a hidden leak that is not the cause, and a collapse on an unseen generator. A generator-agnostic lip-sync stream supplies complementary coverage, and augmentation plus multi-generator training begin to close the generalization gap. The lasting contribution is the evaluation methodology: diagnose leaks, break shortcuts, and always test across generators.

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## Appendix A. Engineering Notes

Key practical issues and resolutions, recorded for reproducibility:

*Table 8. Selected implementation issues and fixes.*

| Issue                                 | Resolution                                       |
|---------------------------------------|--------------------------------------------------|
| ffmpeg not found                      | install via winget; auto-locate in config        |
| transformers rejects .bin (torch<2.6) | load safetensors                                 |
| mediapipe has no solutions (0.10.35)  | use Tasks FaceLandmarker API                     |
| sync timestamp error (VIDEO mode)     | use IMAGE mode (per-frame)                       |
| Coqui dependency clash                | isolate TTS in a separate venv; pin transformers |